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TERMINAL MODULE FOR ROTARY ELECTRIC MACHINE

Abstract

A terminal module (**100**) electrically connects a stator constituting a rotary electric machine to a terminal block (**80**). The terminal module (**100**) includes a plurality of busbars (**10, 20, 30, 40**) and a holding member (**50**) covering the plurality of busbars (**10, 20, 30, 40**). The plurality of busbars (**10, 20, 30**) respectively have elongated extension portions (**13, 23, 33**) located between first end portions (**11, 21, 31**) and second end portions (**12, 22, 32**) and extending in a first direction (X), and intermediate portions located between the extension portions (**13, 23, 33**) and the second end portions (**12, 22, 32**). The extension portions (**13, 23, 33**) are configured to be capable of reducing vibration transmitted toward the terminal block (**80**) from the stator.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a terminal module for rotary electric machine.

BACKGROUND

[0002] Patent Document 1 discloses a busbar that electrically connects an electric motor to a power supply circuit that supplies power to the electric motor. This busbar has a terminal portion on one end side attached to a terminal block of the power supply circuit, a terminal portion on the other end side attached to a stator terminal of the electric motor, and a main body portion that is joined to both terminal portions and electrically connects the terminal portions.

[0003] The main body portion is constituted by a conductor having flexibility.

[0004] According to such a busbar, vibration transmitted from the electric motor is absorbed by the flexible main body portion.

PRIOR ART DOCUMENT

Patent Document

[0005] Patent Document 1: JP 2021-009781 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0006] Incidentally, in a terminal module for rotary electric machine equipped with such a busbar, the main body portion and terminal portions of the busbar are provided separately, and thus the number of components increases, and time and effort is needed to join the main body portion and the terminal portions.

[0007] An object of the present disclosure is to provide a terminal module for rotary electric machine that is able to reduce vibration transmitted from a rotary electric machine with a simple configuration.

Means to Solve the Problem

[0008] A terminal module for rotary electric machine of the present disclosure is a terminal module for rotary electric machine that electrically connects a stator constituting a rotary electric machine to a terminal block, including a plurality of busbars each integrally formed of a metal plate material, and a holding member formed of an electrically insulating resin, covering the plurality of busbars, and interposed between the plurality of busbars, whereby, when an axial direction and a circumferential direction of the rotary electric machine are respectively defined as a first direction and a second direction, the plurality of busbars include a first busbar, a second busbar, and a third busbar disposed in alignment with each other in the second direction and each having a first end portion to be electrically connected to the terminal block, a second end portion to be electrically connected to a coil of the stator, an elongated extension portion located between the first end portion and the second end portion and extending in the first direction, and an intermediate portion located between the extension portion and the second end portion and covered by the holding member, and the extension portion is configured to be capable of reducing vibration transmitted toward the terminal block from the stator.

Effect of the Invention

[0009] According to the present disclosure, vibration transmitted from a rotary electric machine can be reduced with a simple configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of one embodiment of a terminal module for rotary electric

machine showing the terminal module attached to a terminal block.

[0011] FIG. 2 is a plan view showing a state in which the terminal module of FIG. 1 is attached to a stator.

[0012] FIG. 3 is a bottom view showing the terminal module of FIG. 1.

[0013] FIG. 4 is a perspective view showing a first busbar and a second busbar.

[0014] FIG. 5 is a perspective view showing a third busbar.

[0015] FIG. 6 is a perspective view showing a fourth busbar.

[0016] FIG. 7 is a perspective view showing a primary mold body.

[0017] FIG. 8 is a bottom view showing the primary mold body of FIG. 7.

[0018] FIG. 9 is a cross-sectional view showing a molding process of the primary mold body of FIG. 7, in a state where a cavity of a first shaping mold is filled with molten resin.

[0019] FIG. 10 is a perspective view showing the primary mold body, the third busbar, and the fourth busbar separated from each other.

[0020] FIG. 11 is a plan view showing a state in which the third busbar and the fourth busbar are attached to the primary mold body.

[0021] FIG. 12 is a bottom view showing a state in which the third busbar and the fourth busbar are attached to the primary mold body.

[0022] FIG. 13 is a cross-sectional view showing a molding process of a secondary mold body, in a state where a cavity of a second shaping mold is filled with molten resin.

[0023] FIG. 14 is a cross-sectional view taken along line 14X-14X in FIG. 2.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Description of Embodiments of Disclosure

[0024] Initially, modes of the present disclosure will be enumerated and described.

[0025] [1] A terminal module for rotary electric machine of the present disclosure is: a terminal module for rotary electric machine that electrically connects a stator constituting a rotary electric machine to a terminal block, including a plurality of busbars each integrally formed of a metal plate material, and a holding member formed of an electrically insulating resin, covering the plurality of busbars, and interposed between the plurality of busbars, whereby, when an axial direction and a circumferential direction of the rotary electric machine are respectively defined as a first direction and a second direction, the plurality of busbars include a first busbar, a second busbar, and a third busbar disposed in alignment with each other in the second direction and each having a first end portion to be electrically connected to the terminal block, a second end portion to be electrically connected to a coil of the stator, an elongated extension portion located between the first end portion and the second end portion and extending in the first direction, and an intermediate portion located between the extension portion and the second end portion and covered by the holding member, and the extension portion is configured to be capable of reducing vibration transmitted toward the terminal block from the stator.

[0026] According to this configuration, the holding member integrally holds the first busbar, the second busbar, and the third busbar, and is interposed between the busbars. Accordingly, the busbars are electrically insulated from each other by the holding member.

[0027] Also, according to the above configuration, the first busbar, the second busbar, and the third busbar have elongated extension portions that extend in the first direction, thus enhancing flexibility. As a result, vibration transmitted toward the terminal block from the stator becomes more likely to be absorbed by the extension portions. Accordingly, vibration transmitted from the rotary electric machine can be reduced with a simple configuration.

[0028] [2] Preferably, terminal module for rotary electric machine further includes a coupling member formed of an electrically insulating resin and integrally covering the extension portion of each of the first busbar, the second busbar, and the third busbar.

[0029] With the busbars, the extension portions are more likely to relatively shift in position, the longer the length of the extension portions in the first direction. In this regard, according to the

above configuration, the extension portions of the busbars are integrally coupled together by the coupling member. Position shift of the extension portions of the busbars from the normal position can thereby be suppressed. Accordingly, work efficiency when connecting the first end portions of the busbars to the terminal block can be improved.

[0030] [3] Preferably, the coupling member has a plurality of holes that the extension portions respectively pass through in the first direction, and the coupling member and the extension portion of at least one of the first busbar, the second busbar, and the third busbar include a restricting part that restricts movement of the coupling member in the first direction relative to the extension portion by an engaging relationship.

[0031] According to this configuration, movement of the coupling member in the first direction relative to the extension portions of the busbars is restricted by the restricting part. Accordingly, positioning of the coupling member with respect to the busbar can be readily performed.

[0032] [4] Preferably, the restricting part includes a raised portion protruding toward an inner circumferential surface of the hole from the extension portion of at least one of the first busbar, the second busbar, and the third busbar, and a recessed portion formed in the inner circumferential surface.

[0033] According to this configuration, any movement by the coupling member in the first direction relative to the extension portion of each busbar is restricted by the engaging relationship between the raised portion protruding from the extension portion and the recessed portion formed in the inner circumferential surface of the hole in the coupling member. Accordingly, the restricting part can be realized with a simple configuration.

[0034] [5] Preferably, the raised portion is provided at a position closer to the first end portion than a central portion of the extension portion in the first direction.

[0035] With the extension portions, position shift from the normal position is more likely to occur closer to the first end portions. In this regard, according to the above configuration, the raised portion is provided at a position closer to the first end portion than the central portion of the extension portion in the first direction. Thus, the coupling member will be located closer to the first end portion than the central portion. Position shift of the extension portions of the busbars from the normal position can thereby be effectively suppressed. Accordingly, work efficiency when connecting the first end portions of the busbars to the terminal block can be further improved.

[0036] [6] Preferably, the intermediate portion of each of the first busbar, the second busbar, and the third busbar has an elongated main body portion extending in the second direction, and a leg portion formed to bend from the main body portion and extend in the first direction on an opposite side to the extension portion and coupled to the second end portion.

[0037] In a conventional terminal module, a leg portion that joins to a second end portion extends in a third direction, that is, in a radial direction of the rotary electric machine, and thus reducing the size of the rotary electric machine in the third direction is difficult.

[0038] In this regard, according to the above configuration, the leg portion of the intermediate portion is formed to bend from the main body portion and extend in the first direction on the opposite side to the first end portion. Thus, the length of the intermediate portion in the third direction can be reduced, compared to the conventional busbar in which the leg portion extends from the main body portion in the third direction. Accordingly, the physical size of the terminal module can be reduced in the third direction.

[0039] [7] Preferably, the holding member has a first holding part covering the first busbar and the second busbar and interposed between the first busbar and the second busbar, and a second holding part covering the first holding part and the third busbar.

[0040] When the cross-sectional area of the gap between the busbars is small in the case where the holding member integrally holds the first busbar, the second busbar, and the third busbar, molten resin will not flow readily through this gap within the cavity formed by the busbars and the shaping mold that forms the holding member. In the case where the cross-sectional area of the flow path on

the opposite side to the above gap with the busbars interposed therebetween within the cavity is large compared to the cross-sectional area of the gap, a difference thereby occurs in the flow pressure of the molten resin that flows through the gap and the flow pressure of the molten resin that flows through the flow path. Thus, there is a problem in that the busbars are likely to shift in position from the normal position due to such a pressure difference.

[0041] In this regard, according to the above configuration, first, a primary mold body constituted by the first busbar, the second busbar and the first holding part is formed, by filling the cavity of the first shaping mold with the molten resin, in a state where the first busbar and the second busbar are inserted into the first shaping mold. Next, a secondary mold body constituted by the primary mold body, the third busbar, and the second holding part is formed, by filling the cavity of the second shaping mold with molten resin, in a state where the primary mold body and the third busbar are inserted into the second shaping mold.

[0042] Thus, the spacing between the first busbar and the second busbar can be increased, compared to the case where the cavity of the shaping mold is filled with molten resin in a state where the first busbar, the second busbar, and the third busbar are inserted into the shaping mold. The occurrence of a difference in flow pressure of the molten resin is thereby suppressed. Accordingly, position shift of the first busbar and the second busbar can be suppressed.

[0043] [8] Preferably, when a radial direction of the rotary electric machine is defined as a third direction, the third busbar has a bent portion between the extension portion and the intermediate portion of the third busbar, and the bent portion has a portion formed to bend from the extension portion and extend on an inner side in the third direction.

[0044] According to this configuration, the height of the extension portion in the first direction, that is, the position of the first end portion in the first direction, and the position of the extension portion in the third direction are adjusted simply by changing the length of the bent portion in the third direction. The dimensions of the extension portion of the third busbar can thereby be adjusted after forming the primary mold body. Accordingly, the third busbar is unlikely to shift in position, compared to the case where the cavity of the shaping mold is filled with molten resin in a state where the first busbar, the second busbar, and the third busbar are inserted into the shaping mold. Accordingly, position shift of the third busbar can be suppressed.

[0045] [9] Preferably, the first holding part has a groove portion into which a portion of the third busbar fits, the groove portion includes a bottom surface that the portion comes into contact with, and a pair of side surfaces that extend upward from both sides of the bottom surface in the second direction, and the pair of side surfaces sandwich the portion in the second direction.

[0046] According to this configuration, when forming the secondary mold body, movement of the third busbar in the second direction relative to the primary mold body due to the flow pressure of the molten resin is restricted by interference between the pair of side surfaces of the groove portion and a portion of the third busbar fitted into the groove. Accordingly, position shift of the third busbar can be further suppressed.

[0047] [10] Preferably, the plurality of busbars includes a fourth busbar to be electrically connected to a neutral wire of the coil, the first holding part includes a base portion having a first contact surface that one end surface of the third busbar in the first direction comes into contact with, and a second contact surface located on an opposite side to the first contact surface in the first direction and that the fourth busbar comes into contact with, and the second holding part covers the fourth busbar.

[0048] According to this configuration, a secondary mold body constituted by the primary mold body, the third busbar, the fourth busbar, and the second holding part is formed, by filling cavity of the second shaping mold with molten resin, in a state where the primary mold body, the third busbar, and the fourth busbar are inserted into the second shaping mold. At this time, the fourth busbar is in contact with the first holding part, and thus there is no gap between the fourth busbar and the first holding part. Position shift of the fourth busbar can thus be suppressed, compared to

the case where the cavity is filled with molten resin in a state where there is a gap between the fourth busbar and the first holding part.

DETAILED DESCRIPTION OF EMBODIMENTS OF DISCLOSURE

[0049] A specific example of a terminal module for rotary electric machine of the present disclosure will be described below with reference to the drawings. In the diagrams, parts of the configuration may be exaggerated or simplified for the convenience of description. Also, the dimensional ratios of various portions may differ between diagrams. Note that the present disclosure is not limited to these illustrative examples and is indicated by the claims, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein. Herein, “orthogonal” is not only strictly orthogonal but also includes generally orthogonal within a range that achieves the operation and effect of the present embodiment.

(Overall Configuration of Terminal Module **100**)

[0050] As shown in FIGS. **1** and **2**, the terminal module **100** electrically connects a rotary electric machine **90** such as a motor generator (electric generator) in a hybrid vehicle or an electric vehicle, for example, and a terminal block **80**.

[0051] The rotary electric machine **90** is, for example, driven by a multi-phase (in the present embodiment, 3-phase consisting of phase U, phase V, phase W) alternating current.

[0052] The rotary electric machine **90** is constituted by a tubular stator **91** provided with a stator core **92** in which a plurality of slots (not shown) are formed and a coil **93** inserted into the slots and a rotor (not shown) disposed on the inner side of the stator **91** in the radial direction. The coil **93** is constituted by three phase coils respectively corresponding to the three phases (phase U, phase V, and phase W).

[0053] The terminal block **80** has a first terminal **81**, a second terminal **82**, and a third terminal **83**, and a housing **84** that accommodates the terminals **81**, **82**, and **83**.

[0054] One end (**81a**, **82a**, **83a**) of each of the terminals **81**, **82**, and **83** in the extension direction is electrically connected to the terminal module **100**. The other end of each of the terminals **81**, **82**, and **83** in the extension direction is electrically connected to an inverter (all not shown) that converts direct current from a battery into alternating current (in the present embodiment, 3-phase alternating current), for example. Note that, in the present embodiment, the first terminal **81** is a terminal corresponding to phase U, the second terminal **82** is a terminal corresponding to phase V, and the third terminal **83** is a terminal corresponding to phase W.

[0055] The terminal module **100** is disposed at one end portion of the stator **91** in the axial direction.

[0056] The terminal module **100** is provided with a first busbar **10**, a second busbar **20**, a third busbar **30**, and a fourth busbar **40** that are electrically connected to the coil **93**, and a holding member **50** that covers the busbars **10**, **20**, **30**, and **40**.

[0057] Also, the terminal module **100** has a coupling member **70** that integrally covers the busbars **10**, **20**, and **30**.

[0058] Hereinafter, each constituent element of the terminal module **100** will be described in detail.

[0059] Note that, henceforth, the axial direction, circumferential direction, and radial direction of the rotary electric machine **90** will be described as a first direction X, a second direction Y, and a third direction Z, respectively. Also, in the first direction X, the side on which the terminal module **100** is disposed with respect to the stator **91** will be described as one side X1 in the first direction, and the opposite side thereto will be described as other side X2 in the first direction. Also, in the third direction Z, the central axis side of the rotary electric machine **90** will be described as inner side Z1 in the third direction, and the opposite side thereto will be described as outer side Z2 in the third direction.

(First Busbar **10**, Second Busbar **20**)

[0060] As shown in FIG. **4**, the first busbar **10** and the second busbar **20** are each integrally formed of a conductive metal plate material.

[0061] The first busbar **10** and the second busbar **20** are aligned at an interval from each other in a second direction Y.

[0062] Note that, henceforth, in the second direction Y, the second busbar **20** side will be described as one side Y1 in the second direction with respect to the first busbar **10**, and the opposite side thereto will be described as other side Y2 in the second direction.

[0063] The first busbar **10** has a first end portion **11** that is electrically connected to the first terminal **81** of the terminal block **80** and a second end portion **12** that is electrically connected to the coil **93** of the stator **91**.

[0064] Also, the first busbar **10** has an extension portion **13** located between the first end portion **11** and the second end portion **12** and an intermediate portion **14** located between the extension portion **13** and the second end portion **12**.

[0065] The first end portion **11** and the extension portion **13** are a flat plate shape that extends in the first direction X as a whole and in a direction orthogonal to both the first direction X and the third direction Z.

[0066] As shown in FIGS. **1** and **4**, the first end portion **11** is provided with a fastening hole **11a** (see FIG. **4**) that passes therethrough in the third direction Z and through which a bolt (not shown) is inserted and is also provided with a nut **11b** (see FIG. **1**). The nut **11b** is fixed to the end surface of the first end portion **11** on the inner side Z1 in the third direction.

[0067] The first end portion **11** and the first terminal **81** (see FIG. **1**) of the terminal block **80** are fixed to each other, by inserting a bolt (not shown) through a fastening hole (not shown) formed in the one end **81a** and the fastening hole **11a** and threading the bolt into the nut **11b**.

[0068] As shown in FIG. **4**, the extension portion **13** is an elongated shape that extends continuously from the first end portion **11** in the first direction X.

[0069] The extension portion **13** has a raised portion **13a**. The raised portion **13a** protrudes from the edge surface of the extension portion **13** on the one side Y1 in the second direction. The raised portion **13a** is provided at a position closer to the first end portion **11** than a central portion of the extension portion **13** in the first direction X.

[0070] The intermediate portion **14** has an intermediate portion main body **14a** that is formed to bend from the extension portion **13** and extend in the second direction Y and a leg portion **14b** that is formed to bend from the intermediate portion main body **14a** and extend in the first direction X and is coupled to the second end portion **12**. The intermediate portion main body **14a** corresponds to the elongated main body portion.

[0071] The intermediate portion main body **14a** is a flat plate shape that extends in both the second direction Y and the third direction Z.

[0072] The leg portion **14b** is coupled to the end surface of the intermediate portion main body **14a** on the inner side Z1 in the third direction. The leg portion **14b** extends from the intermediate portion main body **14a** on the opposite side to the extension portion **13** in the first direction X, that is, on the other side X2 in the first direction.

[0073] The second end portion **12** is formed to bend from the end portion of the leg portion **14b** on the other side X2 in the first direction and protrude toward the inner side Z1 in the third direction as a whole.

[0074] The second end portion **12** has a flat plate-shaped flat plate portion **12a** located on the leg portion **14b** side in the third direction Z and a rod-shaped protruding portion **12b** located on the distal end side thereof.

[0075] The flat plate portion **12a** is provided with a through hole **15** that passes therethrough in the first direction X.

[0076] The protruding portion **12b** is an elongated shape that extends in the third direction Z. The protruding portion **12b** protrudes from the end surface of the flat plate portion **12a** on the inner side Z1 in the third direction. Note that, in the present embodiment, the protruding portion **12b** is a portion that is connected to a power line (not shown) of the phase coil of the coil **93** corresponding

to phase U.

[0077] In the configuration in which the first busbar **10** includes the elongated extension portion **13**, the total length of the first end portion **11** and the extension portion **13** in the first direction X can be made larger than the total length of the second end portion **12** and the intermediate portion **14** in the first direction X. With this configuration, the flexibility of the first busbar **10** is improved, enabling the effect of absorbing vibration with the first busbar **10** (extension portion **13**) to be enhanced.

[0078] As shown in FIG. 4, the second busbar **20** has a first end portion **21** that is electrically connected to the third terminal **83** of the terminal block **80** and a second end portion **22** that is electrically connected to the coil **93** of the stator **91**.

[0079] Also, the second busbar **20** has an extension portion **23** located between the first end portion **21** and the second end portion **22** and an intermediate portion **24** located between the extension portion **23** and the second end portion **22**.

[0080] The first end portion **21** and the extension portion **23** are flat plate shapes that extend in the first direction X as a whole and in a direction orthogonal to both the first direction X and the third direction Z.

[0081] As shown in FIGS. 1 and 4, the first end portion **21** is provided with a fastening hole **21a** (see FIG. 4) that passes therethrough in the third direction Z and through which a bolt (not shown) is inserted and is also provided with a nut **21b** (see FIG. 1). The nut **21b** is fixed to the end surface of the first end portion **21** on the inner side Z1 in the third direction.

[0082] The first end portion **21** and the third terminal **83** (see FIG. 1) of the terminal block **80** are fixed to each other, by inserting a bolt (not shown) through a fastening hole (not shown) formed in the one end **83a** and the fastening hole **21a** and threading the bolt into the nut **21b**.

[0083] As shown in FIG. 4, the extension portion **23** is an elongated shape that extends continuously from the first end portion **21** in the first direction X.

[0084] The extension portion **23** has a raised portion **23a**. The raised portion **23a** protrudes from the edge surface of the extension portion **23** on the one side Y1 in the second direction. The raised portion **23a** is provided at a position closer to the first end portion **21** than a central portion of the extension portion **23** in the first direction X.

[0085] The intermediate portion **24** has an intermediate portion main body **24a** that is formed to bend from the extension portion **23** and extend in the second direction Y and a leg portion **24b** that is formed to bend from the intermediate portion main body **24a** and extend in the first direction X and is coupled to the second end portion **22**. The intermediate portion main body **24a** corresponds to the elongated main body portion.

[0086] The intermediate portion main body **24a** is a flat plate shape that extends in both the second direction Y and the third direction Z.

[0087] The leg portion **24b** is coupled to the end surface of the intermediate portion main body **24a** on the inner side Z1 in the third direction. The leg portion **24b** extends from the intermediate portion main body **24a** on the opposite side to the extension portion **23** in the first direction X, that is, on the other side X2 in the first direction.

[0088] The second end portion **22** is formed to bend from the end portion of the leg portion **24b** on the other side X2 in the first direction and protrude toward the inner side Z1 in the third direction as a whole.

[0089] The second end portion **22** has a flat plate-shaped flat plate portion **22a** located on the leg portion **24b** side in the third direction Z and a rod-shaped protruding portion **22b** located on the distal end side thereof.

[0090] The flat plate portion **22a** is provided with a through hole **25** that passes therethrough in the first direction X.

[0091] The protruding portion **22b** is an elongated shape that extends in the third direction Z. The protruding portion **22b** protrudes from the end surface of the flat plate portion **22a** on the inner side

Z1 in the third direction. Note that, in the present embodiment, the protruding portion **22b** is a portion that is connected to a power line (not shown) of the phase coil of the coil **93** corresponding to phase W.

[0092] In the configuration in which the second busbar **20** includes the elongated extension portion **23**, the total length of the first end portion **21** and the extension portion **23** in the first direction X can be made larger than the total length of the second end portion **22** and the intermediate portion **24** in the first direction X. With this configuration, the flexibility of the second busbar **20** is improved, enabling the effect of absorbing vibration with the second busbar **20** (extension portion **23**) to be enhanced.

(Third Busbar **30**)

[0093] As shown in FIG. 5, the third busbar **30** is integrally formed of a conductive metal plate material.

[0094] The third busbar **30** has a first end portion **31** that is electrically connected to the second terminal **82** of the terminal block **80** and a second end portion **32** that is electrically connected to the coil **93** of the stator **91**.

[0095] Also, the third busbar **30** has an extension portion **33** located between the first end portion **31** and the second end portion **32**, an intermediate portion **34** located between the extension portion **33** and the second end portion **32**, and a bent portion **35** located between the extension portion **33** and the intermediate portion **34**.

[0096] The first end portion **31** and the extension portion **33** are flat plate shapes that extend in the first direction X as a whole and in a direction orthogonal to both the first direction X and the third direction Z.

[0097] As shown in FIGS. 1 and 5, the first end portion **31** is provided with a fastening hole **31a** (see FIG. 5) that passes therethrough the third direction Z and through which a bolt (not shown) is inserted and is also provided with a nut **31b** (see FIG. 1). The nut **31b** is fixed to the end surface of the first end portion **31** on the inner side **Z1** in the third direction.

[0098] The first end portion **31** and the second terminal **82** (see FIG. 1) of the terminal block **80** are fixed to each other, by inserting a bolt (not shown) into a fastening hole (not shown) formed in the one end **82a** and the fastening hole **31a** and threading the bolt into the nut **31b**.

[0099] As shown in FIG. 5, the extension portion **33** is an elongated shape that extends continuously from the first end portion **31** in the first direction X.

[0100] The extension portion **33** has a raised portion **33a**. The raised portion **33a** protrudes from the edge surface of the extension portion **33** on the one side **Y1** in the second direction. The raised portion **33a** is provided at a position closer to the first end portion **31** than a central portion of the extension portion **33** in the first direction X.

[0101] The bent portion **35** includes a first portion **35a** that is formed to bend from the extension portion **33** and extend in the third direction Z and a second portion **35b** that is formed to bend from the first portion **35a** and extend in the first direction X and is coupled to the intermediate portion **34**.

[0102] The first portion **35a** is formed to bend from the extension portion **33** and extend on the inner side **Z1** in the third direction.

[0103] The second portion **35b** is formed to bend from the distal end of the first portion **35a** and extend on the other side **X2** in the first direction.

[0104] The intermediate portion **34** has an elongated intermediate portion main body **34a** that extends in the second direction Y and a leg portion **34b** that is formed to bend from the intermediate portion main body **34a** and extend in the first direction X and is coupled to the second end portion **32**. The intermediate portion main body **34a** corresponds to the elongated main body portion.

[0105] The intermediate portion main body **34a** is a flat plate shape that extends in both the first direction X and the second direction Y as a whole. The second portion **35b** of the bent portion **35** is

coupled to a central portion, in the second direction Y, of the end surface of the intermediate portion main body **34a** on the one side **X1** in the first direction.

[0106] The leg portion **34b** is formed to bend from the end portion of the intermediate portion main body **34a** on the one side **Y1** in the second direction and extends on the opposite side to the extension portion **33** in the first direction X, that is, on the other side **X2** in the first direction.

[0107] The second end portion **32** is formed to bend from the end portion of the leg portion **34b** on the other side **X2** in the first direction and protrude toward the inner side **Z1** in the third direction as a whole.

[0108] The second end portion **32** has a flat plate-shaped flat plate portion **32a** located on the leg portion **34b** side in the third direction Z and a rod-shaped protruding portion **32b** located on the distal end side thereof.

[0109] The flat plate portion **32a** is provided with a through hole **36** that passes therethrough in the first direction X.

[0110] The protruding portion **32b** is an elongated shape that extends in the third direction Z. The protruding portion **32b** protrudes from the end surface of the flat plate portion **32a** on the inner side **Z1** in the third direction. Note that, in the present embodiment, the protruding portion **32b** is a portion that is connected to a power line (not shown) of the phase coil of the coil **93** corresponding to phase V.

[0111] In the configuration in which the third busbar **30** includes the elongated extension portion **33**, the total length of the first end portion **31** and the extension portion **33** in the first direction X can be made longer than the total length of the second end portion **32** and the intermediate portion **34** in the first direction X or the total length of the second end portion **32**, the intermediate portion **34**, and the bent portion **35** in the first direction X. With this configuration, the flexibility of the third busbar **30** is improved, enabling the effect of absorbing vibration with the third busbar **30** (extension portion **33**) to be enhanced.

(Fourth Busbar **40**)

[0112] As shown in FIG. 6, the fourth busbar **40** is integrally formed of a conductive metal plate material.

[0113] The fourth busbar **40** forms a neutral of the coil **93**, and has third end portions **43A**, **43B**, and **43C** that are connected to a neutral wire (not shown) of the coil **93** and an intermediate portion **40A** to which the third end portions **43A**, **43B**, and **43C** are coupled.

[0114] The intermediate portion **40A** extends in the second direction Y as a whole.

[0115] The intermediate portion **40A** has a first intermediate portion **41** located between the third end portion **43A** and the third end portion **43B**, and a second intermediate portion **42** located between the third end portion **43B** and the third end portion **43C**.

[0116] The first intermediate portion **41** is a flat plate shape that extends in both the second direction Y and the third direction Z.

[0117] The second intermediate portion **42** extends upward on the one side **X1** in the first direction from the end portion of the first intermediate portion **41** on the one side **Y1** in the second direction and extends toward the one side **Y1** in the second direction.

[0118] The third end portions **43A**, **43B**, and **43C** are aligned in the stated order from the other side **Y2** in the second direction toward the one side **Y1** in the second direction. The third end portions **43A**, **43B**, and **43C** are aligned equidistantly from each other in the second direction Y.

[0119] The third end portions **43A** and **43B** protrude from the end surface of the first intermediate portion **41** on the inner side **Z1** in the third direction toward the inner side **Z1** in the third direction as a whole.

[0120] The third end portions **43A** and **43B** each have a flat plate-shaped flat plate portion **43a** located on the first intermediate portion **41** side in the third direction Z and a rod-shaped protruding portion **43b** located on the distal end side thereof.

[0121] The flat plate portion **43a** is provided with a through hole **44** that passes therethrough in the

first direction X.

[0122] The protruding portion **43b** is an elongated shape that extends in the third direction Z. The protruding portion **43b** protrudes from the end surface of the flat plate portion **43a** on the inner side **Z1** in the third direction.

[0123] The third end portion **43C** is formed to bend from the end surface of the second intermediate portion **42** on the other side **X2** in the first direction and protrudes toward the inner side **Z1** in the third direction as a whole.

[0124] The third end portion **43C** has an inclined portion **43c** located on the second intermediate portion **42** side in the third direction Z and a rod-shaped protruding portion **43d** located on the distal end side thereof.

[0125] The inclined portion **43c** is inclined so as to proceed toward the one side **Y1** in the second direction, as the distance from the second intermediate portion **42** in the third direction Z increases.

[0126] The protruding portion **43d** is an elongated shape that extends in the third direction Z. The protruding portion **43d** protrudes from the end surface of the inclined portion **43c** on the inner side **Z1** in the third direction.

[0127] Note that, in the present embodiment, the protrusions **43b** and **43d** are portions that are connected to a neutral wire (not shown) of the coil **93**.

(Basic Configuration of Holding Member **50**)

[0128] As shown in FIGS. **1** to **8**, the holding member **50** is formed of a resin material having electrical insulating properties, and covers the busbars **10**, **20**, **30**, and **40**.

[0129] The holding member **50** has a first holding part **51** that covers the first busbar **10** and the second busbar **20** and is interposed between both busbars **10** and **20** and a second holding part **57** that covers the first holding part **51**, the third busbar **30**, and the fourth busbar **40**.

[0130] Also, the holding member **50** has a plurality of communication holes **60** that communicate with the through holes **15**, **25**, **36**, and **44** in the first direction X. The communication holes **60** include first holes **61a** and **61b** formed in the first holding part **51** and second holes **62** formed in the second holding part **57**.

[0131] Hereinafter, constituent elements of the holding member **50** will be described in detail.

(First Holding Part **51**)

[0132] As shown in FIGS. **4**, **7** and **8**, the first holding part **51** has a base portion **51A**, a first enclosing portion **52** that covers the extension portions **13** and **23**, and a second enclosing portion **53** that covers the second end portions **12** and **22**.

[0133] The base portion **51A** extends in the second direction Y and collectively covers the entirety of the intermediate portions **14** and **24** of the first busbar **10** and the second busbar **20**.

[0134] As shown in FIG. **11**, the base portion **51A** has a first contact surface **C1** that the end surface of the intermediate portion **34** of the third busbar **30** on the other side **X2** in the first direction comes into contact with. The first contact surface **C1** extends along the intermediate portion **34**. The first contact surface **C1** is provided on an end surface **51a** of the base portion **51A** on the one side **X1** in the first direction (near side in direction orthogonal to surface of FIG. **11**).

[0135] As shown in FIGS. **8** and **12**, a fitting recessed portion **54A** into which the intermediate portion **40A** of the fourth busbar **40** fits is formed in an end surface **51b** on the opposite side to the end surface **51a** of the base portion **51A** in the first direction X. A bottom surface of the fitting recessed portion **54A** constitutes a second contact surface **C2** that the end surface of the intermediate portion **40A** on the one side **X1** in the first direction comes into contact with.

[0136] As shown in FIGS. **4** and **7**, the first enclosing portion **52** protrudes on the one side **X1** in the first direction from the end surface **51a** of the base portion **51A** and partially covers the extension portions **13** and **23**.

[0137] As shown in FIGS. **7**, **10** and **11**, the first enclosing portion **52** has a first groove portion **55** into which the bent portion **35** of the third busbar **30** fits on the end surface thereof on the one side **X1** in the first direction. The first groove portion **55** is located in a central portion of the first

enclosing portion **52** in the second direction Y.

[0138] The first groove portion **55** has a bottom surface **55a** that the first portion **35a** of the bent portion **35** of the third busbar **30** comes into contact with, and a pair of side surfaces **55b** that extend upward on the one side X1 in the first direction from both sides of the bottom surface **55a** in the second direction Y.

[0139] As shown in FIG. **11**, the pair of side surfaces **55b** sandwich the first portion **35a** of the bent portion **35** in the second direction Y. The distance between the pair of side surfaces **55b** is slightly larger than the width of the first portion **35a**.

[0140] As shown in FIGS. **4**, **7** and **8**, the second enclosing portion **53** protrudes on the inner side Z1 in the third direction from the base portion **51A** and covers the flat plate portions **12a** and **22a** of the second end portions **12** and **22**.

[0141] As shown in FIGS. **8** and **12**, a fitting recessed portion **54B** that joins to the fitting recessed portion **54A** and into which the third end portions **43A** and **43B** of the fourth busbar **40** fit is formed in an end surface **53b** of the second enclosing portion **53** on the other side X2 in the first direction. A bottom surface of the fitting recessed portion **54B** constitutes a third contact surface C3 that the end surfaces of the third end portions **43A** and **43B** on the one side X1 in the first direction come into contact with.

[0142] As shown in FIGS. **4**, **7** and **8**, the second enclosing portion **53** is provided with two each of the first holes **61a** and **61b**. The first holes **61a** and **61b** are provided alternately from the other side Y2 in the second direction toward the one side Y1 in the second direction. The first holes **61a** respectively communicate with the through hole **15** of the first busbar **10** and the through hole **25** of the second busbar **20** in the first direction X.

[0143] As shown in FIGS. **7** and **8**, a second groove portion **56** into which part of the intermediate portion **34** and the second end portion **32** of the third busbar **30** fit is provided in the end surface of the first holding part **51** on the inner side Z1 in the third direction. The second groove portion **56** extends continuously from the base portion **51A** and to the second enclosing portion **53** in the first direction X. The second groove portion **56** has a bottom surface **56a** that the leg portion **34b** of the intermediate portion **34** of the third busbar **30** comes into contact with and a pair of side surfaces **56b** that extend upward on the inner side Z1 in the third direction from both sides of the bottom surface **56a** in the second direction Y.

[0144] As shown in FIGS. **11** and **12**, the pair of side surfaces **56b** sandwich the leg portion **34b** of the intermediate portion **34** and the flat plate portion **32a** of the second end portion **32** in the second direction Y. The distance between the pair of side surfaces **56b** is slightly larger than the width of the leg portion **34b** and the flat plate portion **32a**.

(Configuration of Second Holding Part 57)

[0145] As shown in FIGS. **1** and **3**, the second holding part **57** has a base portion **58** that extends in the second direction Y and a third enclosing portion **59** that covers part of the third busbar **30**.

[0146] As shown in FIGS. **1** to **8**, the base portion **58** covers the flat plate portion **32a** of the second end portion **32** of the third busbar **30**. Also, the base portion **58** covers the end surface of the first holding part **51** on the other side X2 in the first direction and covers the intermediate portion **40A** and parts of the third end portions **43A**, **43B**, and **43C** of the fourth busbar **40**. Specifically, the base portion **58** covers the flat plate portions **43a** of the third end portions **43A** and **43B** and the inclined portion **43c** of the third end portion **43C**.

[0147] As shown in FIG. **3**, five second holes **62** are formed in the base portion **58**.

[0148] As shown in FIGS. **1** and **3** to **8**, the second holes **62** respectively communicate in the first direction X with the through hole **15** of the first busbar **10**, the through hole **25** of the second busbar **20**, the through hole **36** of the third busbar **30**, and the two through holes **44** of the fourth busbar **40**. The second holes **62** communicating with the through holes **15** and **25** also communicate in the first direction X with the first holes **61a**.

[0149] As shown in FIGS. **1** and **5**, the third enclosing portion **59** protrudes on the one side X1 in

the first direction from the base portion **58**. The third enclosing portion **59** covers the intermediate portion **34** and part of the second portion **35b** of the bent portion **35** of the third busbar **30**. Also, the second groove portion **56** of the first holding part **51** is occupied by the third enclosing portion **59** and the base portion **58** (see FIG. 1).

(Coupling Member **70**)

[0150] As shown in FIGS. 1, 2 and 14, the coupling member **70** is formed of a resin material having electrical insulating properties, and integrally covers the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30**.

[0151] The coupling member **70** has a first portion **71** that covers the first busbar **10**, a second portion **72** that covers the third busbar **30**, and a third portion **73** that covers the second busbar **20**. [0152] The second portion **72** is located between the first portion **71** and the third portion **73** in the third direction Z.

[0153] The first portion **71**, the second portion **72**, and the third portion **73** are aligned in the second direction Y, and the portions that are adjacent to each other in the second direction Y are coupled together.

[0154] The first portion **71**, the second portion **72**, and the third portion **73** respectively have holes **71a**, **72a**, and **73a** that the extension portions **13**, **33**, and **23** pass through in the first direction X.

[0155] As shown in FIG. 14, a recessed portion **71c** is formed in an inner circumferential surface **71b** of the hole **71a** at a position corresponding to the raised portion **13a** protruding toward the inner circumferential surface **71b**. Similarly, recessed portions (not shown) are formed in inner circumferential surfaces of the holes **72a** and **73a** at positions respectively corresponding to the raised portions **33a** and **23a** that protrude toward the inner circumferential surfaces. Note that the recessed portions of the second portion **72** and the third portion **73** are provided with a similar configuration to the recessed portion **71c** of the first portion **71**. Thus, henceforth, only the recessed portion **71c** of the first portion **71** will be described, and description of the recessed portions of the second portion **72** and the third portion **73** will be omitted.

[0156] The raised portion **13a** is fitted into the recessed portion **71c**. The recessed portion **71c** and the raised portion **13a** are configured to restrict movement of the first portion **71**, that is, the coupling member **70**, in the first direction X relative to the extension portion **13**, by engaging with each other.

[0157] Note that the raised portion **13a** and the recessed portion **71c** correspond to the restricting part described in the Description of Embodiments of Disclosure section.

(Method for Manufacturing Terminal Module **100**)

[0158] Next, a method for manufacturing the terminal module **100** will be described, with reference to FIGS. 9 to 13. Note that FIG. 9 corresponds to a cross-sectional view taken along line 9X-9X in FIG. 8, and FIG. 13 corresponds to a cross-sectional view taken along line 13X-13X in FIG. 11.

[0159] First, as shown in FIG. 9, a positioning pin **114** is inserted into the through hole **25** of the second busbar **20**. Also, although not shown, a positioning pin **114** is inserted into the through hole **15** of the first busbar **10**. In this state, an upper mold **111** and a lower mold **112** of a first shaping mold **110** are clamped. The first busbar **10** and the second busbar **20** inserted into the first shaping mold **110** are thereby positioned with respect to a cavity **113**.

[0160] Next, the cavity **113** of the first shaping mold **110** is filled with molten resin R1. A primary mold body **101** constituted by the first busbar **10**, the second busbar **20**, and the first holding part **51** is thereby formed. At this time, the first holes **61a** that respectively communicate with the through holes **15** and **25** are formed in the first holding part **51**. Also, the first holes **61b** are formed in the first holding part **51** by pins not shown.

[0161] Next, as shown in FIG. 10, the third busbar **30** and the fourth busbar **40** are attached to the primary mold body **101**.

[0162] As shown in FIGS. 10 and 11, the first portion **35a** of the bent portion **35** is fitted into the first groove portion **55**, and the leg portion **34b** of the intermediate portion **34** and the flat plate

portion **32a** of the second end portion **32** are fitted into the second groove portion **56**. The third busbar **30** is thereby positioned with respect to the primary mold body **101**.

[0163] At this time, the end surface of the intermediate portion main body **34a** of the third busbar **30** on the other side **X2** in the first direction is in contact with the end surface **51a** (first contact surface **C1**) of the base portion **51A** (see FIG. **13**). Also, the first portion **35a** of the bent portion **35** is in contact with the bottom surface **55a** of the first groove portion **55** (see FIG. **1**). Also, the end surface of the leg portion **34b** on the outer side **Z2** in the third direction is in contact with the bottom surface **56a**.

[0164] Here, the length of the bent portion **35** in the third direction **Z** may be adjusted. Specifically, the length of the first portion **35a** of the bent portion **35** in the third direction **Z** is changed. The height of the extension portion **33** in the first direction **X**, that is, the position of the first end portion **31** in the first direction **X**, and the position of the extension portion **33** in the third direction **Z** are thereby adjusted.

[0165] Note that the position of the first end portion **31** in the first direction **X** is adjusted in the same manner as the positions of the first end portions **11** and **21** in the first direction **X** (see FIG. **1**). Also, the position of the extension portion **33** in the third direction **Z** is adjusted such that the extension portion **33** is located between the extension portion **13** of the first busbar **10** and the extension portion **23** of the second busbar **20** in the third direction **Z** (see FIG. **11**). The extension portions **13**, **23**, and **33**, and, hence, the first end portions **11**, **21**, and **31**, will thereby be aligned in the second direction **Y**.

[0166] As shown in FIGS. **10** and **12**, the fourth busbar **40** is attached to the fitting recessed portions **54A** and **54B** of the first holding part **51**. The fourth busbar **40** is thereby positioned in the normal attachment position with respect to the primary mold body **101**. At this time, the end surface of the fourth busbar **40** on the one side **X1** in the first direction will be in contact with the second contact surface **C2** of the base portion **51A** and the third contact surface **C3** of the second enclosing portion **53**.

[0167] Note that, as shown in FIG. **13**, in a state where the third busbar **30** and the fourth busbar **40** are attached to the primary mold body **101**, the third busbar **30**, the second busbar **20**, and the fourth busbar **40** are disposed at an interval from each other in the first direction **X**. Also, similarly, the third busbar **30**, the first busbar **10** (not shown), and the fourth busbar **40** are disposed at an interval from each other in the first direction **X**.

[0168] Next, a positioning pin **124** is inserted into the through hole **25** of the second busbar **20**. Also, although not shown, positioning pins **124** are inserted into the through holes **36** and **44** of the third busbar **30** and the fourth busbar **40**. In this state, as shown in FIG. **13**, an upper mold **121** and a lower mold **122** of a second shaping mold **120** are clamped. The primary mold body **101**, the third busbar **30**, and the fourth busbar **40** inserted into the second shaping mold **120** are thereby positioned with respect to a cavity **123**.

[0169] Next, the cavity **123** of the second shaping mold **120** is filled with molten resin **R2**. A secondary mold body **102** constituted by the primary mold body **101**, the third busbar **30**, the fourth busbar **40**, and the second holding part **57** is thereby formed. At this time, as shown in FIG. **13**, the second holes **62** that communicate with the first holes **61a** communicating with the through hole **15** and the through hole **25**, the through hole **36**, and the first holes **61b** communicating with the two through holes **44** are formed in the second holding part **57**.

[0170] Note that, at this time, the second end portion **12**, the third end portion **43A**, the second end portion **22**, the third end portion **43B**, the second end portion **32**, and the third end portion **43C** are disposed in the stated order from the other side **Y2** in the second direction toward the one side **Y1** in the second direction and are aligned equidistantly from each other in the second direction **Y** (see FIGS. **1** to **3**).

[0171] Next, the operation and effect of the present embodiment will be described.

[0172] (1) The first busbar **10** is located between the first end portion **11** and the second end portion

12, and has the elongated extension portion **13** that extends in the first direction X. The second busbar **20** is located between the first end portion **21** and the second end portion **22**, and has the elongated extension portion **23** that extends in the first direction X. The third busbar **30** is located between the first end portion **31** and the second end portion **32**, and has the elongated extension portion **33** that extends in the first direction X. The extension portions **13**, **23**, and **33** are configured to be capable of reducing vibration transmitted toward the terminal block **80** from the stator **91**.

[0173] According to this configuration, the holding member **50** integrally holds the first busbar **10**, the second busbar **20**, and the third busbar **30**, and is interposed between the busbars **10**, **20**, and **30**. Accordingly, the busbars **10**, **20**, and **30** are electrically insulated from each other by the holding member **50**.

[0174] Also, according to the above configuration, the first busbar **10**, the second busbar **20**, and the third busbar **30** have elongated extension portions **13**, **23**, and **33** that extend in the first direction X, and thus flexibility is enhanced. As a result, vibrations transmitted toward the terminal block **80** from the stator **91** becomes more likely to be absorbed by the extension portions **13**, **23**, and **33**. Accordingly, vibration transmitted from the rotary electric machine **90** can be reduced with a simple configuration.

[0175] (2) The coupling member **70** that is formed of an electrically insulating resin and integrally covers the extension portions **13**, **23**, and **33** of the first busbar **10**, the second busbar **20**, and the third busbar **30** is provided.

[0176] With the busbars **10**, **20**, and **30**, the extension portions **13**, **23**, and **33** are more likely to relatively shift in position, the longer the length of the extension portions **13**, **23**, and **33** in the first direction X. In this regard, according to the above configuration, the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30** are integrally coupled together by the coupling member **70**.

Position shift of the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30** from the normal position can thereby be suppressed. Accordingly, work efficiency when connecting the first end portions **11**, **21**, and **31** of the busbars **10**, **20**, and **30** to the terminal block **80** can be improved.

[0177] (3) The coupling member **70** has the plurality of holes **71a**, **72a**, and **73a** that pass through the extension portions **13**, **23**, and **33** in the first direction X. The extension portion **13** of the first busbar **10** has the raised portion **13a** serving as a restricting part. The coupling member **70** has the recessed portion **71c** serving as a restricting part.

[0178] According to this configuration, movement of the coupling member **70** in the first direction X relative to the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30** is restricted by the restricting part. Accordingly, positioning of the coupling member **70** with respect to the busbars **10**, **20**, and **30** can be readily performed.

[0179] (4) The restricting part has the raised portion **13a** protruding toward the inner circumferential surface **71b** of the hole **71a** from the extension portion **13** of the first busbar **10** and the recessed portion **71c** formed on the inner circumferential surface **71b**.

[0180] According to this configuration, any movement by the coupling member **70** in the first direction X relative to the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30** is restricted by the engaging relationship between the raised portion **13a** protruding from the extension portion **13** and the recessed portion **71c** formed on the inner circumferential surface **71b** of the hole **71a** in the coupling member **70**. Accordingly, the restricting part can be realized with a simple configuration.

[0181] (5) The raised portion **13a** (**23a**, **33a**) is provided at a position closer to the first end portion **11** (**21**, **31**) than the central portion of the extension portion **13** (**23**, **33**) in the first direction X.

[0182] With the extension portions **13**, **23**, and **33**, position shift from the normal position is more likely to occur closer to the first end portions **11**, **21**, and **31**. In this regard, according to the above configuration, the raised portion **13a** is provided at a position closer to the first end portion **11** than the central portion of the extension portion **13** in the first direction X. Thus, the coupling member

70 will be located closer to the first end portion **11** than the central portion. Position shift of the extension portions **13**, **23**, and **33** of the busbars **10**, **20**, and **30** from the normal position can thereby be effectively suppressed. Accordingly, work efficiency when connecting the first end portions **11**, **21**, and **31** of the busbars to the terminal block **80** can be further improved.

[0183] (6) The intermediate portions **14**, **24**, and **34** of the first busbar **10**, the second busbar **20**, and the third busbar **30** have the elongated intermediate portions **14a**, **24a**, and **34a** that extend in the second direction Y and the leg portions **14b**, **24b**, and **34b** that are formed to bend from the intermediate portion main bodies **14a**, **24a**, and **34a** and extend on the opposite side to the extension portions **13**, **23**, and **33** in the first direction X and are coupled to the second end portions **12**, **22**, and **32**.

[0184] In a conventional terminal module, leg portions that join to the second end portions **12**, **22**, and **32** extend in the third direction Z, and thus reducing the size of the rotary electric machine **90** in the third direction Z is difficult.

[0185] In this regard, according to the above configuration, the leg portions **14b**, **24b**, and **34b** of the intermediate portions **14**, **24**, and **34** are bent from the intermediate portion main bodies **14a**, **24a**, and **34a** and extend on the opposite side to the first end portions **11**, **21**, and **31** in the first direction X. Thus, the lengths of the intermediate portions **14**, **24**, and **34** in the third direction Z can be reduced, compared to the conventional busbars in which the leg portions **14b**, **24b**, and **34b** extend from the intermediate portion main bodies **14a**, **24a**, and **34a** in the third direction Z.

Accordingly, the physical size of the terminal module **100** can be reduced in the third direction Z.

[0186] (7) The holding member **50** has the first holding part **51** that covers the first busbar **10** and the second busbar **20** and is interposed between the first busbar **10** and the second busbar **20**, and the second holding part **57** that covers the first holding part **51** and the third busbar **30**.

[0187] When the cross-sectional area of the gap between the busbars **10**, **20**, and **30** is small in the case where the holding member **50** integrally holds the first busbar **10**, the second busbar **20**, and the third busbar **30**, molten resin will not flow readily through this gap within the cavity formed by the busbars **10**, **20**, and **30** and the shaping mold that forms the holding member **50**. In the case where the cross-sectional area of the flow path on the opposite side to the above gap with the busbar **10**, **20**, and **30** interposed therebetween within the cavity is large compared to the cross-sectional area of the gap, a difference thereby occurs between the flow pressure of the molten resin that flows through the gap and the flow pressure of the molten resin that flows through the flow path. Thus, there is a problem in that the busbars **10**, **20**, and **30** are likely to shift in position from the normal position due to such a pressure difference.

[0188] In this regard, according to the above configuration, first, the primary mold body **101** constituted by the first busbar **10**, the second busbar **20** and the first holding part **51** is formed, by filling the cavity **113** of the first shaping mold **110** with the molten resin R1, in a state where the first busbar **10** and the second busbar **20** are inserted into the first shaping mold **110**. Next, the secondary mold body **102** constituted by the primary mold body **101**, the third busbar **30**, and the second holding part **57** is formed, by filling the cavity **123** of the second shaping mold **120** with the molten resin R2, in a state where the primary mold body **101** and the third busbar **30** are inserted into the second shaping mold **120**.

[0189] Thus, the interval between the first busbar **10** and the second busbar **20** can be increased, compared to the case where the cavity of the shaping mold is filled with molten resin in a state where the first busbar **10**, the second busbar **20**, and the third busbar **30** are inserted into the shaping mold. The occurrence of a difference in flow pressure of the molten resin R1 is thereby suppressed. Accordingly, position shift of the first busbar **10** and the second busbar **20** can be suppressed.

[0190] (8) The third busbar **30** has the bent portion **35** between the extension portion **33** and the intermediate portion **34** of the third busbar **30**. The bent portion **35** has the first portion **35a** that is formed to bend from the extension portion **33** and extend on the inner side Z1 in the third direction.

[0191] According to this configuration, the height of the extension portion **33** in the first direction X, that is, the position of the first end portion **31** in the first direction X, and the position of the extension portion **33** in the third direction Z are adjusted simply by changing the length of the bent portion **35** in the third direction Z. The dimensions of the extension portion **33** of the third busbar **30** can thereby be adjusted after forming the primary mold body **101**. Accordingly, the third busbar **30** is less likely to shift in position, compared to the case where the cavity of the shaping mold is filled with molten resin in a state where the first busbar **10**, the second busbar **20**, and the third busbar **30** are inserted into the shaping mold. Accordingly, position shift of the third busbar **30** can be suppressed.

[0192] (9) The first holding part **51** has the first groove portion **55** into which the first portion **35a** of the bent portion **35** of the third busbar **30** fits. The first groove portion **55** includes the bottom surface **55a** that the first portion **35a** comes into contact with and the pair of side surfaces **55b** that extend upward from both sides of the bottom surface **55a** in the second direction Y. The pair of side surfaces **55b** sandwich the first portion **35a** in the second direction Y. Also, the first holding part **51** has the second groove portion **56** into which the leg portion **34b** and the flat plate portion **32a** of the third busbar **30** fit. The second groove portion **56** includes the bottom surface **56a** that the leg portion **34b** comes into contact with and the pair of side surfaces **56b** that extend upward from both sides of the bottom surface **56a** in the second direction Y. The pair of side surfaces **56b** sandwich the leg portion **34b** and the flat plate portion **32a** in the second direction Y.

[0193] According to this configuration, when forming the secondary mold body **102**, movement of the third busbar **30** in the second direction Y relative to the primary mold body **101** due to the flow pressure of the molten resin R2 is restricted by interference between the pair of side surfaces **55b** of the first groove portion **55** and the first portion **35a** of the third busbar **30** fitted into the first groove portion **55**. Similarly, movement of the third busbar **30** in the second direction Y relative to the primary mold body **101** is restricted by interference between the pair of side surfaces **56b** of the second groove portion **56** and the leg portion **34b** and the flat plate portion **32a** of the third busbar **30** fitted into the second groove portion **56**. Accordingly, position shift of the third busbar **30** can be further suppressed.

[0194] (10) The plurality of busbars have the fourth busbar **40** that is electrically connected to a neutral wire of the coil **93**. The first holding part **51** is provided with the base portion **51A** having the first contact surface C1 that one end surface of the third busbar **30** in the first direction X comes into contact with and the second contact surface C2 and the third contact surface C3 that are located on the opposite side to the first contact surface C1 in the first direction X and that the fourth busbar **40** comes into contact with. The second holding part **57** covers the fourth busbar **40**.

[0195] According to this configuration, the secondary mold body **102** constituted by the primary mold body **101**, the third busbar **30**, the fourth busbar **40** and the second holding part **57** is formed, by filling the cavity **123** of the second shaping mold **120** with the molten resin R2, in a state where the primary mold body **101**, the third busbar **30**, and the fourth busbar **40** are inserted into the second shaping mold **120**. At this time, the fourth busbar **40** is in contact with the first holding part **51**, and thus there is no gap between the fourth busbar **40** and the first holding part **51**. Position shift of the fourth busbar **40** can thus be suppressed, compared to the case where the cavity is filled with the molten resin R2 in a state where there is a gap between the fourth busbar **40** and the first holding part **51**.

<Modifications>

[0196] The present embodiment can be implemented in a modified manner as follows. The present embodiment and the following modifications can be implemented in combination with each other to the extent that no technical inconsistencies arise.

[0197] The terminal module **100** may have a temperature sensor that detects the temperature of the coil **93** by detecting the temperature of the busbars electrically connected to the coil **93**. In this case, the disposition of the temperature sensor can be selected as appropriate according to the

requirements for installing the terminal module **100** on the stator **91**. For example, the temperature sensor may be provided at the end portion of the terminal module **100** on the one side **Y1** in the second direction or may be provided at the end portion on the other side **Y2** in the second direction. [0198] The shapes of the busbars **10**, **20**, **30**, and **40** are not limited to those illustrated in the present embodiment and may be modified as appropriate according to the requirements for installation on the stator **91**, within a range that achieves the operation and effect of the present disclosure. For example, the second intermediate portion **42** of the fourth busbar **40** need not extend upward from the end portion of the first intermediate portion **41** on the one side **Y1** in the second direction, and may be a flat plate shape that extends in both the second direction **Y** and the third direction **Z**.

[0199] The number of fourth busbars **40** is not limited to one as illustrated in the present embodiment, and may, for example, be two or more. Also, the fourth busbar **40** may be omitted. In this case, the fitting recessed portions **54A** and **54B** can be omitted from the first holding part **51**.

[0200] The first holding part **51** may, for example, have holes formed by a jig or a positioning pin, in addition to the first holes **61a** and **61b**. Also, similarly, the second holding part **57** may, for example, have holes formed by a jig or a positioning pin, in addition to the second holes **62**.

[0201] The second holding part **57** is not limited to having the second holes **62** that communicate with the first holes **61a** and **61b** in the first direction **X** as illustrated in the present embodiment. That is, the second holes **62** that communicate with the first holes **61a** and **61b** in the first direction **X** may be omitted from the second holding part **57**.

[0202] The number and disposition of the through holes **15**, **25**, **36**, and **44** are not limited to those illustrated in the present embodiment and may be changed as appropriate according to the shape of the busbars **10**, **20**, **30**, and **40**.

[0203] The communication holes **60** are not limited to having the second holes **62**. That is, the second holes **62** may be omitted from the second holding part **57**. In this case, the through hole **36** can be omitted from the third busbar **30**. Also, in this case, when forming the secondary mold body **102**, for example, the third busbar **30** need only be sandwiched by a jig from both sides in the first direction **X**.

[0204] The first holes **61a** and **61b** may be omitted. In this case, the through holes **15** and **25** can be omitted from the first busbar **10** and the second busbar **20**. Also, in this case, when forming the primary mold body **101**, the first busbar **10** and the second busbar **20** need only be sandwiched by a jig from both sides in the first direction **X**, for example.

[0205] The first groove portion **55** may be omitted from the first holding part **51**.

[0206] The second groove portion **56** may be omitted from the first holding part **51**.

[0207] The bent portion **35** may be omitted from the third busbar **30**. Even in this case, in the third busbar **30**, the extension portion **33** need only be located between the extension portion **13** of the first busbar **10** and the extension portion **23** of the second busbar **20** in the second direction **Y**.

[0208] The holding member **50** is not limited to having the first holding part **51** and the second holding part **57**. That is, in the terminal module **100**, the busbars may be collectively held by the holding member **50**.

[0209] In the busbars **10**, **20**, and **30**, the leg portions **14b**, **24b**, and **34b** of the intermediate portions **14**, **24**, and **34** need not be formed to bend from the intermediate portion main bodies **14a**, **24a**, and **34a** and extend on the opposite side to the extension portions **13**, **23**, and **33** in the first direction **X**. For example, the leg portions **14b**, **24b**, and **34b** may be formed to bend from the intermediate portion main bodies **14a**, **24a**, and **34a** and extend on the inner side **Z1** in the third direction.

[0210] The raised portions **13a**, **23a**, and **33a** are not limited to those provided at positions closer to the first end portions **11**, **21**, and **31** than the central portion of the extension portions **13**, **23**, and **33** in the first direction **X**. For example, the raised portions **13a**, **23a**, and **33a** may be provided in the central portion, or may be provided at position closer to the intermediate portions **14**, **24**, and **34**

than the central portion in the first direction X.

[0211] The number and disposition of the raised portions **13a**, **23a**, and **33a** may be changed as follows. For example, any one of the raised portions **13a**, **23a**, and **33a** may be omitted, or any two of the raised portions **13a**, **23a**, and **33a** may be omitted. Also, in the present embodiment, the raised portions **13a**, **23a**, and **33a** are illustrated as protruding from the end surfaces of the extension portions **13**, **23**, and **33** on the one side **Y1** in the second direction, but may be modified to protrude from the end surfaces on the other side **Y2** in the second direction, or may be modified to protrude one each from the end surfaces on both sides in the second direction **Y**.

[0212] The raised portions **13a**, **23a**, and **33a** may be omitted. In this case, the recessed portions can be omitted from the holes **71a**, **72a**, and **73a** of the coupling member **70**.

[0213] The coupling member **70** may be omitted.

[0214] The disposition of the plurality of busbars is not limited to that illustrated in the present embodiment and can be changed as appropriate according to the requirements for installation on the stator **91**. Even in this case, the plurality of busbars need only be disposed at an interval from each other in the first direction **X**, and need only be disposed in alignment with each other in the second direction **Y**.

[0215] The present disclosure encompasses the following embodiments. Note that, reference numerals indicating constituent elements of the above embodiment are given merely for the purpose of aiding understanding and are not limiting.

Supplementary Note 1

[0216] A terminal module (**100**) for rotary electric machine that electrically connects a stator (**91**) constituting a rotary electric machine to a terminal block (**80**), including: a plurality of busbars (**10**, **20**, **30**, **40**) each integrally formed of a metal plate material; and a holding member (**50**) formed of an electrically insulating resin, covering the plurality of busbars, and interposed between the plurality of busbars, wherein, when an axial direction and a circumferential direction of the rotary electric machine are respectively defined as a first direction (**X**) and a second direction (**Y**), the plurality of busbars (**10**, **20**, **30**, **40**) include a first busbar (**10**), a second busbar (**20**), and a third busbar (**30**) disposed in alignment with each other in the second direction (**Y**), the first, second, and third busbars each have: a first end portion (**11**, **21**, **31**) to be electrically connected to the terminal block (**80**); a second end portion (**12**, **22**, **32**) to be electrically connected to a coil (**93**) of the stator (**91**); an elongated extension portion (**13**, **23**, **33**) located between the first end portion and the second end portion and extending in the first direction (**X**); and an intermediate portion (**14**, **24**, **34**) located between the extension portion and the second end portion and covered by the holding member (**50**), and a total length of the first end portion and the extension portion in the first direction (**X**) is greater than a total length of the second end portion and the intermediate portion in the first direction (**X**).

LIST OF REFERENCE NUMERALS

[0217] **C1** First abutting surface [0218] **C2** Second abutting surface [0219] **C3** Third abutting surface [0220] **R1**, **R2** Molten resin [0221] **X** First direction [0222] **X1** One side [0223] **X2** Other side [0224] **Y** Second direction [0225] **Y1** One side [0226] **Y2** Other side [0227] **Z** Third direction [0228] **Z1** Inner side [0229] **Z2** Outer side [0230] First busbar [0231] **11** First end portion [0232] **11a** Fastening hole [0233] **11b** Nut [0234] **12** Second end portion [0235] **12a** Flat plate portion [0236] **12b** Protruding portion [0237] **13** Extension portion [0238] **13a** Raised portion [0239] **14** Intermediate portion [0240] **14a** Intermediate portion main body [0241] **14b** Leg portion [0242] **15** Through hole [0243] **20** Second busbar [0244] **21** First end portion [0245] **21a** Fastening hole [0246] **21b** Nut [0247] **22** Second end portion [0248] **22a** Flat plate portion [0249] **22b** Protruding portion [0250] **23** Extension portion [0251] **23a** Raised portion [0252] **24** Intermediate portion [0253] **24a** Intermediate portion main body [0254] **24b** Leg portion [0255] **25** Through hole [0256] **30** Third busbar [0257] **31** First end portion [0258] **31a** Fastening hole [0259] **31b** Nut [0260] **32** Second end portion [0261] **32a** Flat plate portion [0262] **32b** Protruding portion [0263] **33**

Extension portion [0264] **33a** Raised portion [0265] **34** Intermediate portion [0266] **34a** Intermediate portion main body [0267] **34b** Leg portion [0268] **35** Bent portion [0269] **35a** First portion [0270] **35b** Second portion [0271] **36** Through hole [0272] **40** Fourth busbar [0273] **40A** Intermediate portion [0274] **41** First intermediate portion [0275] **42** Second intermediate portion [0276] **43A**, **43B**, **43C** Third end portion [0277] **43a** Flat plate portion [0278] **43b** Protruding portion [0279] **43c** Inclined portion [0280] **43d** Protruding portion [0281] **44** Through hole [0282] **50** Holding member [0283] **51** First holding part [0284] **51A** Base portion [0285] **51a** End surface [0286] **51b** End surface [0287] **52** First enclosing portion [0288] **53** Second enclosing portion [0289] **53b** End surface [0290] **54A**, **54B** Fitting recessed portion [0291] **55** First groove portion [0292] **55a** Bottom surface [0293] **55b** Side surface [0294] **56** Second groove portion [0295] **56a** Bottom surface [0296] **56b** Side surface [0297] **57** Second holding part [0298] **58** Base portion [0299] **59** Third enclosing portion [0300] **60** Communication hole [0301] **61a**, **61b** First hole [0302] **62** Second hole [0303] **70** Coupling member [0304] **71** First portion [0305] **71a** Hole [0306] **71b** Inner circumferential surface [0307] **71c** Recessed portion [0308] **72** Second portion [0309] **72a** Hole [0310] **73** Third portion [0311] **73a** Hole [0312] **80** Terminal block [0313] **81** First terminal [0314] **81a** One end [0315] **82** Second terminal [0316] **82a** One end [0317] **83** Third terminal [0318] **83a** One end [0319] **84** Housing [0320] **90** Rotary electric machine [0321] **91** Stator [0322] **92** Stator core [0323] **93** Coil [0324] **100** Terminal module [0325] **101** Primary mold body [0326] **102** Secondary mold body [0327] **110** First shaping mold [0328] **111** Upper mold [0329] **112** Lower mold [0330] **113** Cavity [0331] **114** Pin [0332] **120** Second mold [0333] **121** Upper mold [0334] **122** Lower mold [0335] **123** Cavity [0336] **124** Pin

Claims

1. A terminal module for rotary electric machine that electrically connects a stator constituting a rotary electric machine to a terminal block, comprising: a plurality of busbars each integrally formed of a metal plate material; and a holding member formed of an electrically insulating resin, covering the plurality of busbars, and interposed between the plurality of busbars, wherein, when an axial direction and a circumferential direction of the rotary electric machine are respectively defined as a first direction and a second direction, the plurality of busbars include a first busbar, a second busbar, and a third busbar disposed in alignment with each other in the second direction and each having: a first end portion to be electrically connected to the terminal block; a second end portion to be electrically connected to a coil of the stator; an elongated extension portion located between the first end portion and the second end portion and extending in the first direction; and an intermediate portion located between the extension portion and the second end portion and covered by the holding member, and the extension portion is configured to be capable of reducing vibration transmitted toward the terminal block from the stator.
2. The terminal module for rotary electric machine according to claim 1, further comprising: a coupling member formed of an electrically insulating resin and integrally covering the extension portion of each of the first busbar, the second busbar, and the third busbar.
3. The terminal module for rotary electric machine according to claim 2, wherein the coupling member has a plurality of holes that the extension portions respectively pass through in the first direction, and the coupling member and the extension portion of at least one of the first busbar, the second busbar, and the third busbar include a restricting part that restricts movement of the coupling member in the first direction relative to the extension portion by an engaging relationship.
4. The terminal module for rotary electric machine according to claim 3, wherein the restricting part includes: a raised portion protruding toward an inner circumferential surface of the hole from the extension portion of at least one of the first busbar, the second busbar, and the third busbar, and a recessed portion formed in the inner circumferential surface.
5. The terminal module for rotary electric machine according to claim 4, wherein the raised portion

is provided at a position closer to the first end portion than a central portion of the extension portion in the first direction.

6. The terminal module for rotary electric machine according to claim 1, wherein the intermediate portion of each of the first busbar, the second busbar, and the third busbar has: an elongated main body portion extending in the second direction; and a leg portion formed to bend from the main body portion and extend in the first direction on an opposite side to the extension portion and coupled to the second end portion.

7. The terminal module for rotary electric machine according to claim 6, wherein the holding member has: a first holding part covering the first busbar and the second busbar and interposed between the first busbar and the second busbar; and a second holding part covering the first holding part and the third busbar.

8. The terminal module for rotary electric machine according to claim 7, wherein, when a radial direction of the rotary electric machine is defined as a third direction, the third busbar has a bent portion between the extension portion and the intermediate portion of the third busbar, and the bent portion has a portion formed to bend from the extension portion and extend on an inner side in the third direction.

9. The terminal module for rotary electric machine according to claim 7, wherein the first holding part has a groove portion into which a portion of the third busbar fits, the groove portion includes: a bottom surface that the portion comes into contact with; and a pair of side surfaces that extend upward from both sides of the bottom surface in the second direction, and the pair of side surfaces sandwich the portion in the second direction.

10. The terminal module for rotary electric machine according to claim 7, wherein the plurality of busbars includes a fourth busbar to be electrically connected to a neutral wire of the coil, the first holding part includes a base portion having: a first contact surface that one end surface of the third busbar in the first direction comes into contact with; and a second contact surface located on an opposite side to the first contact surface in the first direction and that the fourth busbar comes into contact with, and the second holding part covers the fourth busbar.
